Lesson 1: Systems of Measurement, Referents, and Measuring Instruments

Pick a referent close in size to the object, count how many times it fits, then multiply; when you need more precision, add an instrument's coarse and fine readings.

1. The Imperial System of Measurement

Before measuring instruments existed, an unknown length was compared to a familiar object or body part — a **referent**. The need for consistent, repeatable measurements produced formal **systems of measurement**. The **imperial system**, developed in England and still in everyday use in the United States, measures length in inches, feet, yards and miles.

$$1 \text{ foot} = 12 \text{ inches} \qquad 1 \text{ yard} = 3 \text{ feet} \qquad 1 \text{ mile} = 1\,760 \text{ yards}$$

Many imperial units began life as referents:

- The **yard** dates to shortly after the Norman Conquest of 1066 — legend holds that King Henry I decreed it to be the distance from the tip of his nose to the end of his outstretched thumb.
- The **furlong** was used by Saxon farmers before 1066 for how far a team of oxen could plow before resting. It still states the distance of horse races: 8 furlongs = 1 mile.
- The **rod** was a pole about 11 cubits long.
- The **league** had several personal referents, the two most common being the distance a person or a horse could walk in one hour — roughly 3 miles. It is no longer in common use.
- The **chain** is the length of a cricket pitch, about 4 rods, and 10 chains = 1 furlong.
- The **hand** is the width of a clenched fist, 4 inches, and still states the height of a horse or pony.

2. The SI System of Measurement

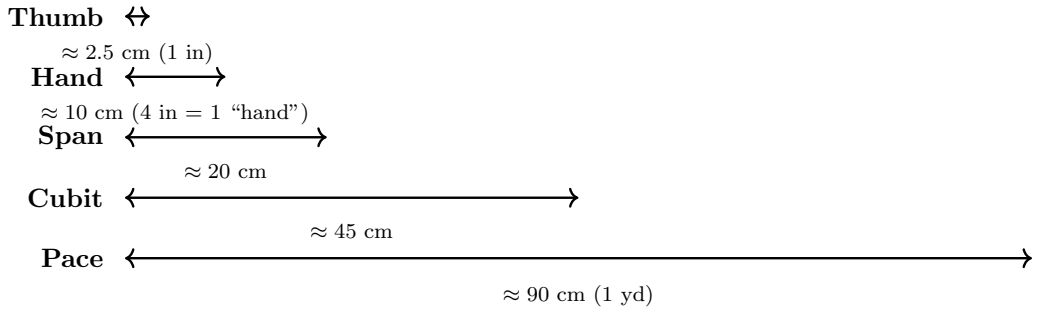
The other major system is the **SI system** (Système International d'Unités), commonly called the **metric system**. As early as 1670 Gabriel Mouton and Jean Picard proposed a base-10 system to replace the tangle of older, inconsistent units, and in 1790 the French Academy of Sciences was asked to design a scientific standard: a unit of length based on a fraction of the Earth's circumference, with every related unit scaled by a power of 10. Canada officially adopted SI in the 1970s, by which time most of the world had already switched. The common SI unit of length is the metre.

$$1 \text{ centimetre} = 10 \text{ millimetres} \qquad 1 \text{ metre} = 100 \text{ centimetres} \qquad 1 \text{ kilometre} = 1\,000 \text{ metres}$$

The United States uses SI units widely in science but still relies on the imperial system for most everyday purposes, so both systems are worth knowing well.

3. Referents: Measuring Without a Tool

A **referent** is a familiar object — very often a part of your own body — used as a stand-in point of reference when no measuring tool is available. A referent taken from a person's immediate surroundings (their own body, or an object they always carry) is a **personal referent**. Body-based referents are the oldest and most convenient, since they are always at hand. The five below are drawn roughly to scale relative to one another.



Example 1 — matching the size of the referent to the size of the object

For each object, state one personal referent (from the diagram above, or from the imperial units it represents) that would be reasonable to use to estimate it, and justify the choice.

(a) the length of a hockey rink **(b)** the length of a USB flash drive **(c)** the height of a pony

4. Referents for Everyday Objects

Connecting a unit to a concrete, familiar referent is the fastest way to build a feel for that unit.

Imperial referents for length:

- the width of a thumb across the joint is about 1 inch.
- the length of an adult foot is about 1 foot.
- one adult pace is about 1 yard.

SI referents for length:

- the width of an adult little ("pinky") fingernail is about 1 centimetre.
- the width of five adult toes together is about 1 decimetre ($\frac{1}{10}$ of a metre).
- the thickness of a dime is about 1 millimetre.
- the average height of a 4-year-old child is about 1 metre.
- the height of a standard doorway is about 2 metres.
- the width of this workbook is about 20 centimetres.
- the diameter of a hockey puck is about 7.6 centimetres.
- the length of an Olympic swimming pool is 50 metres.

Example 2 — one object, one referent from each system

Consider the task of estimating the length of a bath towel.

(a) Identify a referent that could be used to estimate the length in SI units.

(b) Identify a referent that could be used to estimate the length in imperial units.

(c) Explain the process you would use to estimate the length with your imperial referent from part (b).

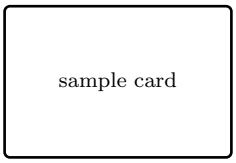
Example 3 — strategies for a perimeter and a circumference

(a) Describe and explain a personal strategy that could be used to estimate the perimeter of the base of a rectangular tool shed, using one of the referents above.

(b) Describe and explain a personal strategy that could be used to estimate the circumference of a bicycle wheel, using one of the referents above.

Example 4 — measuring a scale diagram in both systems

Shown below is a scale diagram of a sample access card, drawn at a scale of 1 : 3. Measure the printed diagram itself — it is reproduced at full size.



(a) Measure the length of the scale diagram above, in both SI units and imperial units.

(b) Use your answer from part (a) to determine the actual length of the access card.

(c) Could the length of this access card be used as a referent in either system of measurement? Explain.

Example 5 — building a big referent out of a smaller one

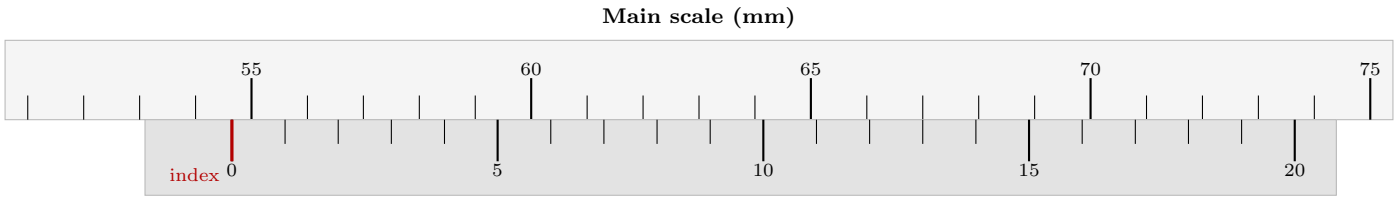
Marcus was asked to give a referent for 1 kilometre. He could not think of anything that was actually 1 kilometre long, so instead he built his referent for the kilometre out of a referent for a smaller unit. Suggest how he might have done this.

5. Measuring Instruments: the Vernier Caliper and the Micrometer

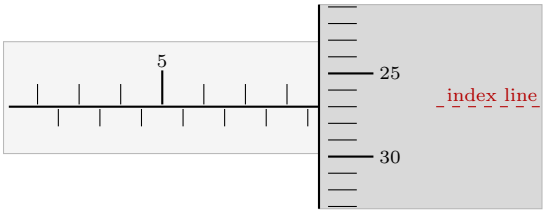
A referent is good to "about". When a job needs more, use an instrument that suits both the **size** of the object and the **precision** required: a **trundle wheel** (rolled along the ground) for long distances, a **tape measure** for rooms, a **ruler** for lengths that fit on a desk (to the nearest millimetre), a **vernier caliper** for small objects to the nearest 0.05 mm (outside and inside dimensions, and the depth of a hole), and a **micrometer** for the thickness of very small objects to the nearest 0.01 mm. A caliper and a micrometer each give a **coarse reading** on a fixed scale and a **fine reading** on a moving scale. Add them, and state the result to the instrument's own precision — 25.30 mm, not 25.3 mm, on a caliper good to 0.05 mm.

Vernier caliper. The coarse reading is the last **main-scale** mark (whole millimetres) at or to the left of the **index**, the vernier's own 0 mark. The fine reading is the **vernier** mark that lines up most closely with a main-scale mark, at 0.05 mm per division.

Micrometer. The coarse reading is the last fully visible **sleeve** mark to the left of the thimble's edge (a mark every 0.5 mm: whole millimetres above the line, halves below it). The fine reading is the **thimble** mark on the index line, at 0.01 mm per division.



The index sits just past the 54 mm mark (coarse reading 54 mm), and vernier mark 13 lines up exactly with a main-scale mark, so the fine reading is $13 \times 0.05 = 0.65$ mm: $54 \text{ mm} + 0.65 \text{ mm} = 54.65$ mm.



The last fully visible sleeve mark is the half mark 8.5 mm (coarse reading). The index line lands on thimble mark 27, so the fine reading is $27 \times 0.01 = 0.27$ mm:

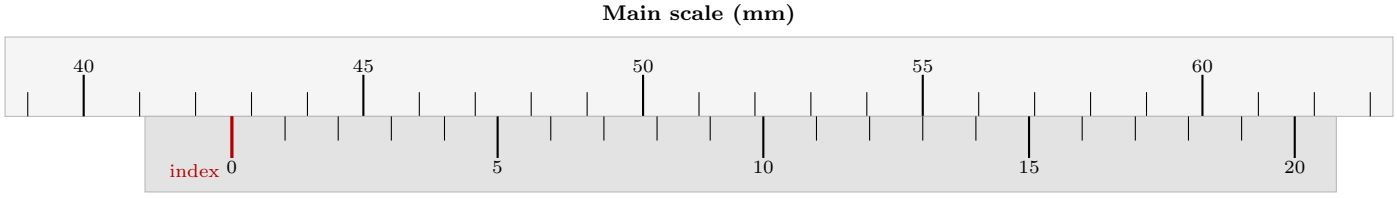
$$8.5 \text{ mm} + 0.27 \text{ mm} = 8.77 \text{ mm}$$

Look below the line: a half mark showing past the last whole millimetre adds 0.5 mm.

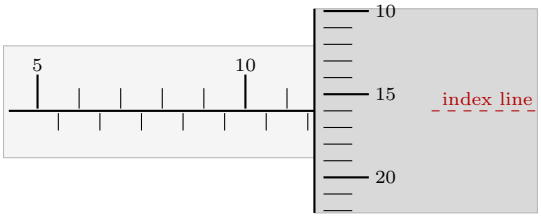
Example 6 — reading a vernier caliper and a micrometer

Determine each reading, in millimetres, to the precision of the instrument.

(a) a vernier caliper



(b) a micrometer



Lesson 2: Converting Units Within and Between SI and Imperial

SI prefixes are powers of ten; imperial units each carry their own multiplier. Estimate first with a benchmark, then convert exactly with a proportion or unit analysis.

Sections 1–3 review conversions *inside* the metric system, where every step is a power of ten. Sections 4–7 bring in the imperial units and the facts that move a measurement between the two systems.

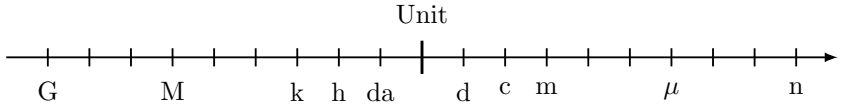
1. SI Review: the Prefix Chart and the Metric Number Line

Every SI measurement is a **base unit** — the metre (length), the gram (mass), or the litre (volume) — carrying a **prefix** that scales it by a power of ten. The chart below gives each common prefix, its symbol, and its value relative to the base unit.

Prefix	giga	mega	kilo	hecto	deca	(base)	deci	centi	milli	micro	nano
Symbol	G	M	k	h	da	m, g, L	d	c	m	μ	n
Value	10^9	10^6	10^3	10^2	10^1	10^0	10^{-1}	10^{-2}	10^{-3}	10^{-6}	10^{-9}

The core prefixes for this unit are **kilo** through **milli**. Giga, mega, micro and nano are **Extension**: notice that each of their steps is a factor of 1000, not 10.

Arranging the prefixes from largest to smallest produces a **metric number line** on which each single step is one power of ten. Moving right (toward smaller prefixes) *multiplies* by 10 per step; moving left (toward larger prefixes) *divides* by 10 per step. Kilometres to millimetres is 6 positions right (k is 3 steps *left* of the unit and m is 3 steps *right* of it), so 1 km $= 10^6$ mm $= 1\,000\,000$ mm; the reverse trip is 6 positions left, so 1 mm $= 10^{-6}$ km.



Example 1 — counting steps on the metric number line

Use the metric number line to convert each measure.
(a) 4.6 m to cm **(b)** 58 400 mg to kg **(c)** 36 mm to m

2. Proportional Reasoning and Map Scales

A **proportion** is a statement that two ratios are equal. Given either a known conversion (such as 1 m = 100 cm) or a ratio supplied by the question, set up $\frac{a}{b} = \frac{c}{d}$ with matching units in matching positions, then cross-multiply to isolate the unknown. Scale drawings, map scales, recipes and currency exchange rates are all ratios, so each can be solved with a proportion.

Map scales. A map scale written as a ratio, $1 : n$, means 1 unit on the map stands for n of the *same* unit on the ground: on a $1 : 50\,000$ map, 1 cm on the map is 50 000 cm = 500 m on the ground. Convert to a convenient unit only at the end.

Example 2 — a map scale written as a ratio, $1 : n$

A map of central Alberta has a scale of $1 : 250\,000$.
(a) Two towns are 18.4 cm apart on the map. How far apart are they on the ground, in kilometres?

(b) Two lakes are 32 km apart. How far apart are they on the map, in centimetres?

3. Unit Analysis (Dimensional Analysis)

Proportional reasoning suits a single conversion. When several conversions must be chained, **unit analysis** (also called **dimensional analysis**) is usually faster, and it is the method used in science courses. The set-up is

$$\text{required} = (\text{given}) \times (\text{conversion}),$$

with each conversion written as a fraction arranged so the unwanted unit cancels. For example, to convert 640 mL to L using 1 L = 1000 mL,

$$\frac{640 \text{ mL}}{1} \times \frac{1 \text{ L}}{1000 \text{ mL}} = \frac{640}{1000} \text{ L} = 0.64 \text{ L} \qquad (\text{mL cancels, leaving L}).$$

A **rate** such as 64 m/s already is a fraction, so each unit that changes (the distance *and* the time) gets its own conversion fraction, arranged to cancel where that unit sits.

Example 3 — chaining two conversion factors, and converting a rate

(a) Complete the unit analysis below to convert 5.3 km to mm. Show the cancellation.

$$\frac{5.3\text{ km}}{1} \times \frac{1000\text{ m}}{1\text{ km}} \times \frac{1000\text{ mm}}{1\text{ m}} =$$

(b) Priya needs to convert 64 m/s to km/min. State the two conversions she needs, then complete her work, showing the cancellation of units.

$$\frac{64\text{ m}}{\text{s}} \times \frac{\text{km}}{\text{m}} \times \frac{\text{s}}{\text{min}} =$$

In Sections 1–3 every SI conversion was a power of ten. Imperial units do not work that way: 12 in make a foot, 3 ft a yard and 5280 ft a mile, and moving between the two systems needs a conversion fact such as 1 in = 2.54 cm. The methods of Sections 1–3 — proportional reasoning and unit analysis — still work; only the conversion facts change.

4. Key Imperial and Imperial–SI Conversion Facts

These are the conversion facts used in the rest of the lesson and in the assignment. Facts marked **(exact)** are true by definition; the others are rounded.

Imperial to Imperial

1 foot (ft) = 12 inches (in) (exact)
1 yard (yd) = 3 feet (ft) (exact)
1 mile (mi) = 5280 ft (exact)
1 mile (mi) = 1760 yards (yd) (exact)

Imperial ↔ SI (Metric)

1 in = 2.54 cm (exact)	1 cm ≈ 0.3937 in
1 ft = 0.3048 m (exact)	1 m ≈ 3.2808 ft
1 yd = 0.9144 m (exact)	1 m ≈ 1.0936 yd
1 mi ≈ 1.6093 km	1 km ≈ 0.6214 mi
1 kg ≈ 2.2 lb (pounds)	

Tip. Use an *exact* factor when you can (for example, divide by 2.54 rather than multiply by 0.3937), so your answer does not differ from the key by one unit in the last place.

Benchmarks for checking (mental math). Before converting, estimate with a round benchmark; after converting, compare. If the two are far apart, look for a multiply/divide mix-up.

1 in ≈ 2.5 cm 1 ft ≈ 30 cm 1 m ≈ 1.1 yd 1 mi ≈ 1.6 km (so 5 mi ≈ 8 km) 1 kg ≈ 2.2 lb

Also check the *direction*: converting to a *smaller* unit gives a *bigger* number (1 ft → about 30 cm), and converting to a *bigger* unit gives a smaller number.

5. Converting Within the Imperial System

Imperial conversions chain the imperial-to-imperial facts exactly the way 1 m = 100 cm is used in SI, except that there are four different multipliers to keep straight (12, 3, 5280, 1760), so write down which fact is being used at each step. Going the other way — from a small unit to a mixed measure — is a division with a remainder.

Example 4 — mixed imperial measures, both directions

Convert each of the following.

(a) 8 ft 5 in to inches

(b) 6 yd 1 ft to feet

(c) 2 mi 410 yd to yards

(d) 68 in to feet and inches

(e) 97 ft to yards and feet

(f) 5555 yd to miles and yards

6. Chaining Conversions with Proportional Reasoning and Unit Analysis

Both methods still work when more than one multiplier is needed. Unit analysis is usually faster, because the units themselves show which factors to multiply by.

Example 5 — the same chained conversion, two ways

Convert 0.6 miles to inches using each method.

Proportional Reasoning **Unit Analysis**

Example 6 — building the conversion fact 1 m ≈ ? in

A student uses unit analysis to find how many inches are in one metre. The required unit is inches per metre. Part of her work is shown.

$$\frac{100 \text{ cm}}{1 \text{ m}} \cdot \frac{1 \text{ in}}{2.54 \text{ cm}} =$$

(a) State which units cancel, and what unit is left.

(b) Calculate the answer to two decimal places.

(c) Write the answer as a completed conversion fact: 1 m ≈ _____ in.

7. Converting Between the Imperial and SI Systems

The imperial–SI facts move a measure directly between systems. When a measurement mixes imperial units (feet *and* inches), convert it to a single imperial unit first, then apply the imperial–SI fact.

Example 7 — across the systems, in both directions

(a) 6 ft 9 in = _____ m (to three decimal places)

(b) 3.15 m = _____ ft _____ in (to the nearest inch)

Example 8 — estimate first, then convert

Use a benchmark to estimate each conversion mentally. Then convert exactly and compare.

(a) 5 ft 9 in to metres (two decimal places)

(b) 40 km to miles (one decimal place)

(c) A student writes 150 lb = 330 kg. Use a benchmark to explain why this cannot be right, then correct it.

Example 9 — athletes' heights, animals' masses

Convert each height to the nearest hundredth of a metre and each mass to the nearest kilogram.
(a) Wayne Gretzky's height, $6'0''$ **(b)** Shaquille O'Neal's height, $7'1''$ **(c)** Simone Biles's height, $4'8''$

(d) a bull moose, about 1200 lb **(e)** a Canada goose, about 9 lb **(f)** a grizzly bear, about 600 lb

Example 10 — a conversion that must be rounded up

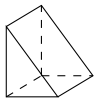
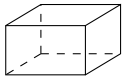
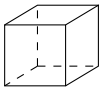
Before digital storage, industrial and educational films were shot on 35 mm film stock, which is now being transferred onto digital media. If 1200 feet of film fit on one archival DVD, determine how many DVDs are needed to store 1650 metres of film footage.

Lesson 3: Surface Area and Volume of Prisms and Cylinders

Volume is base area times height for every right prism and cylinder; surface area is the total area of every face, read off the solid or off its net.

1. Right Prisms

A **prism** has two congruent, parallel bases, with every side face a parallelogram. A **right prism** is a prism whose side faces are all rectangles — each side face meets the base at a right angle. In the three right prisms below the two congruent bases are the front and back ends.

Right Triangular Prism	**Right Rectangular Prism**	**Cube**
		
Two triangular ends, three rectangular sides.	Two rectangular ends, four rectangular sides.	Two square ends, four square sides — six square faces.

Formulas for any right prism.

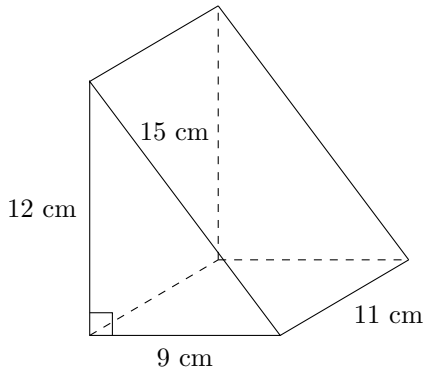
$$V_{\text{Prism}} = A_{\text{Base}} \times h$$

where A_{Base} is the area of one base and h is the distance between the two bases (often called the *length* of a triangular prism). For **surface area**, add together the areas of every face — the two congruent bases plus every rectangular side face.

2. Volume and Surface Area of a Right Triangular Prism

Example 1 — base area, then volume, then every face

Consider the right triangular prism shown, whose triangular base is a right triangle.



(a) Determine the area of the triangular base of the prism.

(b) Determine the volume of the prism.

(c) Determine the surface area of the prism.

3. Working With Volume and Capacity Units

The volume of a solid is measured in cm^3, m^3, in^3, ft^3, and so on. The **capacity** of a container — how much liquid it holds — is measured in litres, millilitres or gallons.

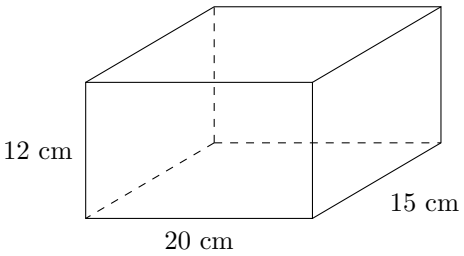
Volume	Solid	Liquid
Metric	cm^3, m^3, km^3, etc.	litres (L), millilitres (mL), etc.
Imperial	in^3, ft^3, yd^3, etc.	gallons, pints, etc.

Conversions. 1 cm^3 = 1 mL 1000 mL = 1 L 1 m^3 = 1000 L 1 US gallon = 3.7854 L 1 imperial (UK) gallon = 4.5461 L 1 gallon = 8 pints

Between systems, *cube* the linear factor of Lesson 2: from 1 ft = 0.3048 m come 1 ft^3 = 0.028 316 8 m^3 and 1 m^3 = 35.3147 ft^3; squaring it gives 1 ft^2 = 0.092 903 m^2.

Example 2 — the same box as a volume and as a capacity

Vault Boxes Inc. manufactures containers in the shape of right rectangular prisms. Nova Blocks Ltd needs a cardboard container 20 cm long, 15 cm wide and 12 cm high to pack 1 cm building blocks. AquaGlow Tanks needs a small glass aquarium with the *same* dimensions, measured on the *inside* of the glass.



(a) Calculate the volume of the container for Nova Blocks Ltd. Is cm^3 an appropriate way to report this volume?

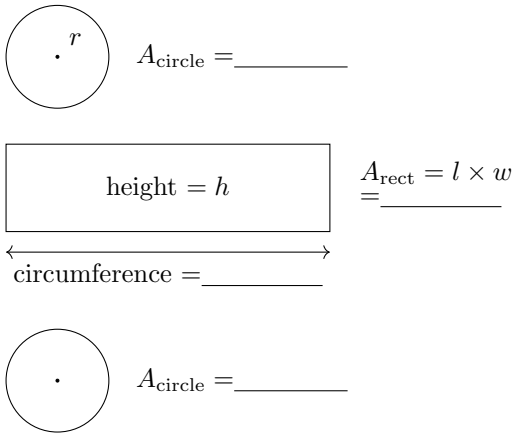
(b) The AquaGlow aquarium will be filled with water, and its capacity is advertised in litres and in US gallons. Determine the capacity of one aquarium in both units, to two decimal places.

4. Volume and Surface Area of a Right Cylinder

A right cylinder behaves exactly like a right prism whose base is a circle instead of a polygon: two congruent, parallel circular bases joined by a curved lateral surface. Unrolling the curved surface gives a flat **net** made of two circles and one rectangle, and that net is the key to the surface area formula.

Example 3 — reading the surface area off the net

The net of a right cylinder with base radius r and height h is shown below: the circular top and bottom, and the rectangle formed by unrolling the curved side.



(a) Fill in the four blanks in the diagram, in terms of r and h.

(b) Use the diagram to show that the surface area of a right cylinder is $SA_{\text{Cylinder}} = 2\pi r^2 + 2\pi rh$.

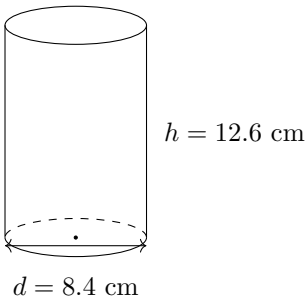
(c) Use the general formula for the volume of a right prism to write the formula for the volume of a cylinder of radius r and height h.

Formulas for a right cylinder.

$$V_{\text{Cylinder}} = \pi r^2 h \qquad SA_{\text{Cylinder}} = 2\pi r^2 + 2\pi rh = 2\pi r(r + h)$$

Example 4 — a canister, from a diameter

A cylindrical storage canister has a height of 12.6 cm and a base diameter of 8.4 cm. Determine the volume and the surface area of the canister, each to the nearest tenth.

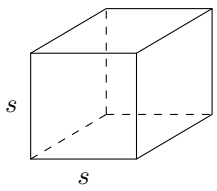


5. Solving for an Unknown Dimension Given the Volume

When the volume is already known and a dimension is asked for, substitute every known value into the appropriate volume formula and solve the resulting equation for the unknown.

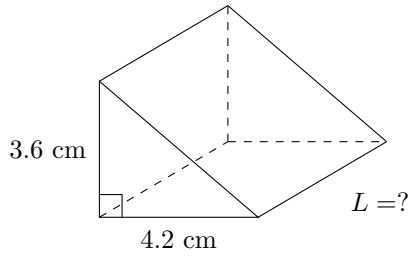
Example 5 — a cube, working backwards from its volume

A commercial recycling bin in the shape of a cube has a volume of 74.5 m^3. Determine the side length s of the bin, to the nearest cm.



Example 6 — a triangular prism, working backwards from its volume

The box for a chocolate bar is shaped like a right triangular prism. The triangular ends have a base of 4.2 cm and a height of 3.6 cm. If the empty space inside the box has a volume of 145.3 cm^3, determine the length L of the box, to the nearest tenth of a cm.

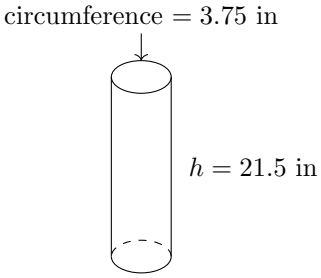


6. Curved Surface Area Only

Some questions ask only for the area of the curved lateral surface of a cylinder — a label wrapped around a can, for instance. That area is the circumference of the base times the height, *without* the area of either circular end.

Example 7 — the label on a poster tube

A shipping company packs its poster tubes in cylindrical cardboard containers with a height of 21.5 in and a base circumference of 3.75 in. Every container is wrapped with a paper label that covers the entire curved surface. Calculate, to the nearest square inch, the area of the label.



7. Solving for an Unknown Dimension Given the Surface Area

Working backwards from a surface area is the same idea as Section 5: write the surface-area formula, substitute everything you know, and solve for the one unknown. For a cylinder, the $2\pi r^2$ part becomes a number, so the height is left in a two-step linear equation: subtract that number, then divide by $2\pi r$.

Example 8 — the height of a can from its surface area

A closed cylindrical can has a surface area of 250 cm^2 and a radius of 3.5 cm. Determine its height, to the nearest tenth of a centimetre.

Lesson 4: Surface Area and Volume of Pyramids and Cones

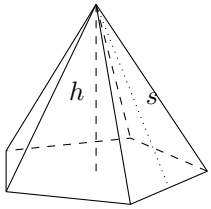
A pyramid or cone holds exactly one third of the prism or cylinder on the same base; surface area uses the slant height s, volume uses the perpendicular height h.

1. Right Pyramids

A **pyramid** has a polygon base and triangular side faces that meet at a single point, the **apex**. A **right pyramid** is a pyramid whose apex is directly above the centre of its base. If the base is a regular polygon, the side faces are congruent isosceles triangles; if the base is a rectangle, opposite side faces match in pairs. Two measurements matter:

- h = the **height** — the perpendicular distance from the apex straight down to the centre of the base.
- s = the **slant height** — the distance from the apex down along a triangular face to the midpoint of a base edge.

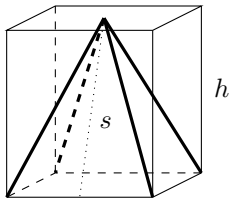
The base can be any polygon (rectangular, triangular, square, pentagonal, hexagonal, ...); the figure shows a regular pentagonal base. Unless stated otherwise, "pyramid" in this course always means *right* pyramid.



2. Volume of a Right Pyramid

In a woodworking class, Mr. Halden's students built a rectangular *planter box* (a right prism) and a pyramid-shaped *funnel* on an identical rectangular base and with the same height. A student filled the funnel with rice and emptied it into the planter box, then repeated this twice more. On the third emptying the box was filled *exactly* to the top — no rice left over, no room left.

The experiment illustrates a general fact: **a pyramid's volume is always exactly $\frac{1}{3}$ of the volume of a prism sharing the same base and the same height** — whatever shape the base is. The figure shows a right pyramid (heavy lines) sharing its base and height with the surrounding prism (light lines).



Question for discussion: How does the volume of a pyramid compare with the volume of a prism that has an identical base and the same height?

Example 1 — square-based pyramid: volume from h, surface area from s

A glass pyramid-shaped skylight sits above a museum atrium. It has a square base with sides measuring 16 m and a height of 6 m.

(a) Make a sketch of the pyramid, labelling the given measurements, and calculate its volume to the nearest cubic metre.

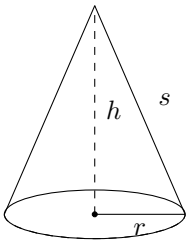
(b) Make a sketch of the net of the pyramid and calculate the surface area to the nearest square metre. Explain why the base is *not* included in this calculation.

3. Right Circular Cones

A **right circular cone** has a circular base and an apex directly above the centre of that base. As with a pyramid, h is the perpendicular height and s is the slant height — here the straight-line distance along the curved surface from the apex to any point on the edge of the base circle. The base radius is r. Since h, r and s are the two legs and the hypotenuse of a right triangle,

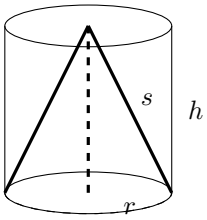
$$s^2 = h^2 + r^2.$$

Unless stated otherwise, "cone" in this course always means *right circular* cone.



4. Volume of a Right Circular Cone

A cone and a cylinder built on an identical circular base with the same height behave exactly as the pyramid and prism did: filling the cone with water and pouring it into the cylinder, three full cones are needed to fill the cylinder exactly. **A cone's volume is always exactly $\frac{1}{3}$ of the volume of a cylinder sharing its base and height** — the same $\frac{1}{3}$ relationship holds for every radius and height.



Question for discussion: How does the volume of a cone compare with the volume of a cylinder that has an identical base and the same height?

5. Formulas for Surface Area and Volume

Surface Area — uses the **slant height** s.

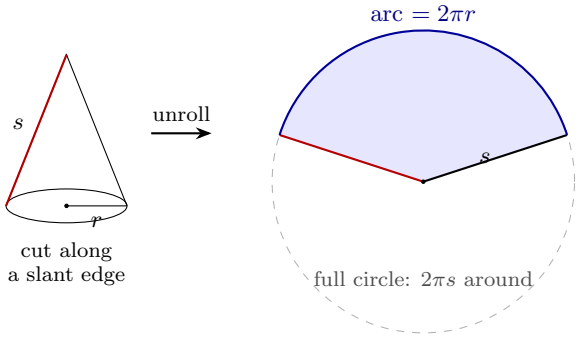
$$SA_{\text{pyramid}} = \text{Area of Base} + \text{Area of Triangular Faces}$$
$$SA_{\text{cone}} = \text{Area of Base} + \text{Area of Curved Surface} = \pi r^2 + \pi rs$$

Volume — uses the **perpendicular height** h.

$$V_{\text{pyramid}} = \tfrac{1}{3}(\text{Volume of prism}) = \tfrac{1}{3}A_{\text{Base}}h$$
$$V_{\text{cone}} = \tfrac{1}{3}(\text{Volume of cylinder}) = \tfrac{1}{3}\pi r^2 h$$

where A_{Base} is the area of the pyramid's base.

Why the curved surface of a cone is πrs. Cut the cone along one slant edge and unroll it. The curved surface flattens into a **sector** (a "pizza slice") of a circle whose radius is the slant height s. The arc of the sector is the edge that wrapped around the base, so its length is the base circumference, $2\pi r$.



The sector is the fraction $\dfrac{\text{arc}}{\text{full circumference}} = \dfrac{2\pi r}{2\pi s} = \dfrac{r}{s}$ of the full circle of radius s, whose area is πs^2. So

$$\text{curved area} = \frac{r}{s} \times \pi s^2 = \pi rs.$$

Adding the circular base gives $SA_{\text{cone}} = \pi r^2 + \pi rs$.

Example 2 — cone: h given, s found by Pythagoras

A solid wooden cone (a closed solid, so its base counts) has a height of 12 cm and a base diameter of 10 cm. Determine the volume and the surface area of the cone, each to one decimal place.

Example 3 — hexagonal base: exact area, then volume

A right pyramid has a height of 12 cm and a regular hexagonal base with sides measuring 6 cm.

(a) Make a sketch of the hexagonal base and determine its area as an exact value (in simplest radical form) and to the nearest square centimetre.

(b) The pyramid is filled completely with water. Determine the volume of water it holds, to the nearest millilitre.

Example 4 — scaling a cone's volume without the formula

Use the information below to answer both parts.

Problem 1: Two cones have radii of equal length. The first cone has a height of 6 cm and a volume of 15 cm^3. State the volume of the second cone if its height is 24 cm.

Problem 2: Two cones have the same height. The first cone has a radius of 4 cm and a volume of 20 cm^3. State the volume of the second cone if its radius is 12 cm.

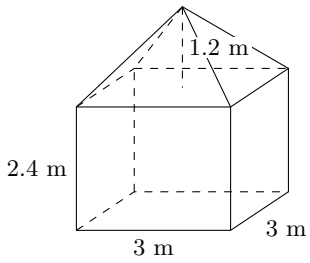
(a) Explain how Problem 1 can be solved using mental math (without the volume formula), and state the answer.

(b) Explain how Problem 2 can be solved using mental math (without the volume formula), and state the answer.

6. Composite Objects

A **composite object** is built from two or more simple solids. Its *volume* is the sum of the volumes of the parts. For its *surface area*, add the outside surfaces of the parts, but leave out any face where two parts are joined: that shared face is hidden inside the object, so it is removed from *both* parts.

Example 5 — a shed: rectangular prism + square pyramid roof



A garden shed is a rectangular prism with a 3 m by 3 m floor and walls 2.4 m high, topped by a right square pyramid roof on the same 3 m by 3 m square, with a height of 1.2 m.

(a) Determine the volume of the shed.

(b) Determine the surface area of the shed (including the floor), to the nearest tenth of a square metre. Which face is *not* counted, and why?

Lesson 5: Surface Area and Volume of Spheres and Hemispheres

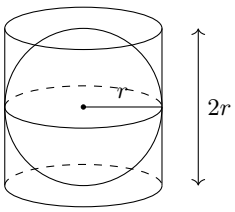
A sphere fills two thirds of the cylinder that just contains it; a hemisphere is half a sphere, and whether its flat circular base is counted is set by the wording of each question.

1. Volume of a Sphere

Mika filled a cylinder with 480 mL of water, balanced it above an empty overflow container, and lowered a solid ball straight down into the cylinder until the water stopped spilling. The ball and the cylinder had the same radius, and the cylinder's height equalled the ball's diameter. Across three trials the overflow container always held exactly 320 mL and exactly 160 mL always stayed in the cylinder. The water pushed out is the water the ball displaces, so the overflow volume *is* the ball's volume; 320 mL is $\frac{2}{3}$ of 480 mL, and the 160 mL left behind is the other $\frac{1}{3}$. So **a sphere's volume is $\frac{2}{3}$ of the volume of a cylinder with the same radius whose height equals the sphere's diameter.**

For discussion. If Mika instead filled the same cylinder with 900 mL of water under the same conditions (cylinder radius = ball radius, cylinder height = ball diameter), how many millilitres would overflow, and how many would remain in the cylinder?

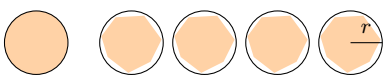
A cylinder that holds a sphere of radius r snugly — the sphere touching the top, the bottom and the curved wall all at once — has radius r and height $2r$. Mika's $\frac{2}{3}$ relationship then turns the cylinder formula $V = \pi r^2 h$ into a formula good for *every* radius, with no need to repeat the experiment:



$$V_{\text{sphere}} = \tfrac{2}{3}V_{\text{cylinder}} = \tfrac{2}{3}\pi r^2 h = \tfrac{2}{3}\pi r^2(2r) = \frac{4}{3}\pi r^3$$

Sphere formulas. $SA_{\text{sphere}} = 4\pi r^2$ $\qquad$ $V_{\text{sphere}} = \dfrac{4}{3}\pi r^3$

Why $4\pi r^2$? The orange-peel test. Trace four circles with the same radius r as an orange. Peel the orange and lay the flattened pieces of peel inside the circles: the peel fills almost exactly *four* circles, so the sphere's surface is $4 \times \pi r^2$. (Archimedes showed the same thing another way: the sphere's surface equals the curved surface of its snug cylinder, $2\pi r \times 2r = 4\pi r^2$.)

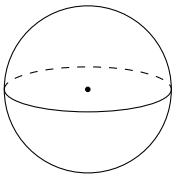


orange

the peel fills about 4 circles of radius r

Example 1 — a spherical tank: halve the diameter first

A spherical propane storage tank has a diameter of 5.8 m.



$d = 5.8$ m

(a) Determine the surface area of the tank, to the nearest tenth of a square metre.

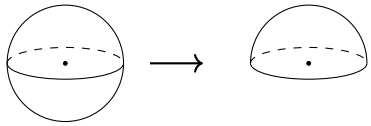
(b) Determine the volume of the tank, to the nearest tenth of a cubic metre.

2. Hemispheres

A **hemisphere** is half of a sphere — the solid produced by slicing a sphere exactly through its centre — just as a **semi-circle** is half of a circle. When a question asks for the surface area of a hemisphere it means **the curved surface plus the flat circular base**, *unless* the base is specifically excluded by the wording (an open bowl, say, or a dome built with no floor).

Example 2 — the three hemisphere formulas, in terms of r and π

A sphere of radius r is sliced exactly in half, giving a hemisphere of the same radius r. Complete each statement below.



sphere, radius r hemisphere, radius r

(a) Volume of a hemisphere = _____

(b) Surface area of a hemisphere = curved surface area + area of circular base
= _____ + _____ = _____

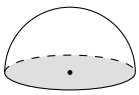
(c) Surface area of a hemisphere, if the base is <u>not</u> included, = _____

Example 3 — a circumference measurement gives the radius

A soccer ball is wrapped once around its widest point with a piece of string, which is then pulled straight and measured with a ruler. The string is 70 cm long. Determine the volume of the soccer ball, to the nearest tenth of a cubic centimetre.

Example 4 — the same dome, with and without its floor

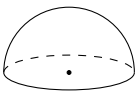
Sana and Tavi are each building a hemisphere-shaped storage shed, both of diameter 12 ft. Sana's shed will include a circular floor built from the same material as the dome; Tavi's shed will have no floor.



$d = 12$ ft

Sana's shed (with floor)

(a) Calculate, in terms of π, how much material Sana needs to build her shed.



$d = 12$ ft

Tavi's shed (no floor)

(b) Calculate, in terms of π, how much material Tavi needs to build his shed.

Example 5 — working backwards from a volume to a radius

A spherical stone sculpture has a volume of 3053.6 ft^3. Determine the radius of the sculpture, to the nearest tenth of a foot.

Lesson 6: Measurement Unit Review

No new formulas: every skill of the unit — conversion, referents, instruments, scale, surface area and volume — assembled into one set of problems.

1. Converting Within the SI System

From kilo through milli (kilo, hecto, deca, *unit*, deci, centi, milli) each step along the SI prefix ladder changes a measurement by a factor of 10; the Extension steps milli → micro → nano are each a factor of 1000, as on the Lesson 2 number line. Either way, every metric conversion is a multiplication or division by a power of ten. Check the direction every time: converting to a *smaller* unit must give a *larger* number.

Example 1 — metric conversions, and judging a set of statements

(a) A rectangular banner measures 54 cm in height and 1.46 m in length. Determine the difference between these two measurements, in millimetres.

(b) *(Extension)* Consider the following four statements. How many of them are false?

Statement 1: 1 cm = 0.01 m Statement 2: 1 mm = 0.001 m Statement 3: 1 μm = 0.01 mm

Statement 4: 1 km = 1000 m

2. Choosing Referents, Units, and Instruments

Choosing a unit means matching the size of the unit to the size of the object; choosing an instrument means matching the precision the job needs to a tool that can deliver it. A **referent** is a familiar object — often a part of the body — whose size is known, used to estimate a measurement with no tool in hand.

Example 2 — units, instruments, and a referent-based calculation

(a) State the most suitable *imperial* unit for measuring each of the following.

the diameter of a coin: _____ the length of a soccer field: _____ the driving distance between two towns: _____

(b) State the measuring instrument that would be easiest to use for each of the following.

the thickness of a credit card: _____ the width of a doorway: _____ the length of a running track: _____

(c) The height of a horse is measured in *hands*. Using the width of an adult hand as a referent for 4 inches, determine the height, in feet, of a horse measuring $14\frac{1}{4}$ hands.

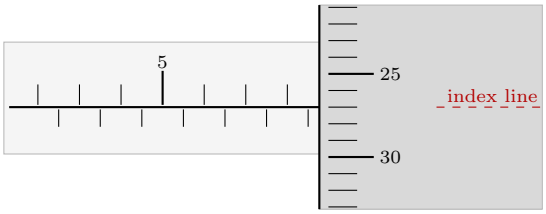
3. Reading a Micrometer

A **micrometer** measures small lengths to a hundredth of a millimetre by combining two scales: a fixed **sleeve scale**, marked in whole and half millimetres, giving the coarse reading; and a rotating **thimble scale**, marked in hundredths of a millimetre, giving the fine reading. The two are added:

$$\text{reading} = (\text{sleeve scale value}) + (\text{thimble scale value}).$$

Example 3 — combining the sleeve and thimble readings

The micrometer below has a measuring range of 0–25 mm and is accurate to 0.01 mm. The sleeve has a mark every 0.5 mm; the thimble has divisions of 0.01 mm. Determine the sleeve (coarse) reading and the thimble (fine) reading separately, then combine them to state the full reading, in millimetres.



Sleeve: a mark every 0.5 **mm**
Thimble: 0.01 **mm per division**

4. Converting Between Systems, and Using Scale

An **approximate conversion** such as 5 miles = 8 km (the Lesson 2 benchmark 1 mi $\approx$ 1.6 km) switches between the systems by a single ratio. A **map scale** written $1 : n$ (Lesson 2, Example 2) means one unit of distance on the map represents n of the same unit in reality, so the map distance is the real distance divided by n — with both expressed in the same unit first.

Example 4 — an approximate conversion and a map scale

(a) A section of highway is 1032 km long. Using the approximate conversion 5 miles = 8 km, determine this distance in miles.

(b) The scale on a map is $1 : 1\,500\,000$. If the actual distance between two towns is 540 km, determine the distance between them on the map, in centimetres.

5. Applied Conversion Problems

Most applied problems give measurements in mixed units. Convert every quantity to the *same* unit before any arithmetic is attempted.

Example 5 — tiling a floor, and dosing a tank

(a) A rectangular hallway floor is 7 yards long and $3\frac{1}{2}$ yards wide. The tiles chosen to cover it are $10\frac{1}{2}$ inches by $10\frac{1}{2}$ inches. Determine the number of tiles needed.

(b) An aquarium has dimensions 55 cm by 40 cm by 25 cm. The directions on a bottle of fish medicine are to add 4 drops per 10 litres of water. Determine the number of drops needed to completely fill the aquarium with treated water.

6. Comparing Imperial and Metric Measurements

Two lengths given in different systems can be compared without a full conversion of both, as long as each is placed on the same number line. Lesson 2's bridge, 1 cm = 0.3937 in, does it in either direction: 1 mm = 0.03937 in, and 1 in = 2.54 cm = 25.4 mm.

Example 6 — choosing a wrench from the other system

Zara received a mixed wrench set. The metric wrenches measure 7 mm, 8 mm, 9 mm, 11 mm, 13 mm, and 16 mm. The imperial wrenches measure $\frac{1}{4}$ in, $\frac{9}{32}$ in, $\frac{3}{8}$ in, $\frac{7}{16}$ in, $\frac{1}{2}$ in, and $\frac{5}{8}$ in.

(a) Zara is tightening a bolt. Her 9 mm wrench is too small and her 11 mm wrench is too large. Which imperial wrench from her set should she try?

(b) On a different bolt, Zara's $\frac{1}{2}$ in wrench is too small and her $\frac{5}{8}$ in wrench is too large. Which metric wrench from her set should she try?

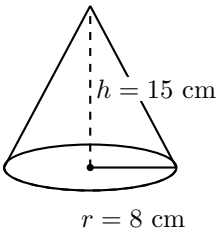
7. Surface Area and Volume Formulas: A Review

Every formula of Lessons 3, 4 and 5 is collected below. Throughout, h is the **perpendicular height**, s is the **slant height**, A_{Base} is the area of the base, and r is the **radius** — halve a diameter before substituting it.

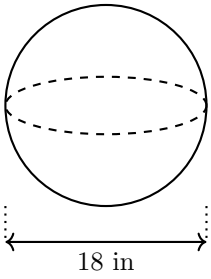
Solid	**Surface area**	**Volume**
Prism	sum of all faces $= 2A_{\text{Base}} + (\text{perimeter})h$	$V = A_{\text{Base}} \times h$
Cylinder	$SA = 2\pi r^2 + 2\pi rh = 2\pi r(r+h)$	$V = \pi r^2 h$
Pyramid	$SA = A_{\text{Base}} + (\text{triangular faces, each } \frac{1}{2}bs)$	$V = \frac{1}{3}A_{\text{Base}}h$
Cone	$SA = \pi r^2 + \pi rs$, where $s^2 = h^2 + r^2$	$V = \frac{1}{3}\pi r^2 h$
Sphere	$SA = 4\pi r^2$	$V = \frac{4}{3}\pi r^3$
Hemisphere	$SA = 3\pi r^2$ with its flat base, $2\pi r^2$ without	$V = \frac{2}{3}\pi r^3$

Example 7 — a cone (slant height first) and a sphere (radius first)

(a) A conical paperweight has a base radius of 8 cm and a height of 15 cm. Determine the surface area of the paperweight, in terms of π.



(b) A spherical balloon has a diameter of 18 inches. Determine its surface area, to the nearest square inch.



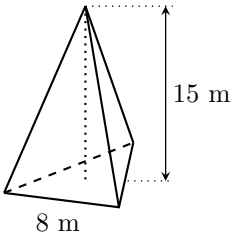
8. Volume of 3-D Solids: A Review

A pyramid or cone holds one third of the prism or cylinder on the same base, so $V = \frac{1}{3}A_{\text{Base}}h$ uses the *perpendicular* height, never the slant height. A volume formula can also be rearranged and solved for a missing dimension: $V = \frac{2}{3}\pi r^3$ gives $r = \sqrt[3]{\frac{3V}{2\pi}}$.

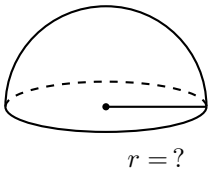
Example 8 — a pyramid's volume, and a hemisphere's radius from its volume

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(a) A sculpture is shaped like a triangular pyramid. Its base is an equilateral triangle with sides of 8 m, and the pyramid's height is 15 m. Determine the volume of the sculpture, as an exact value in simplest radical form and to one decimal place.



(b) A hemispherical storage dome has a volume of 1526.8 m^3. Determine the radius of the dome, to the nearest tenth of a metre.



9. Cumulative Applied Problem

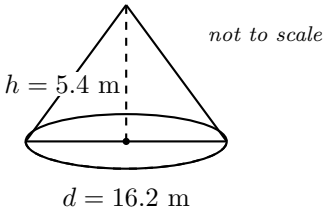
One problem can chain the volume of a cone, the volume of a cylinder, a volume formula solved for a missing dimension, and a cost comparison that needs both a unit conversion and a currency conversion. As everywhere in this unit, carry each value forward at full calculator precision and round only the answer a part asks for.

Example 9 — a conical tank, tanker trucks, a second tank, and two bids

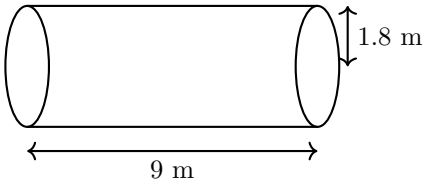
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A specialty fertilizer concentrate is stored in a conical tank in Calgary whose diameter is three times its height. The height of the tank is 5.4 m.

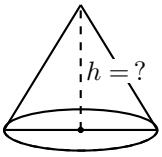
(a) Determine the volume of concentrate the tank holds, correct to one decimal place.



(b) The concentrate is transferred using cylindrical tanker trucks, each with a radius of 1.8 m and a length of 9 m. Using the volume from part (a), determine the number of trucks needed.



(c) At the distributor, the concentrate is poured into a new conical tank with a circular base of diameter 12 m. Using the total volume from part (a), determine the height of this tank, to one decimal place.



(d) Two shipping companies bid on the transport job. Prairie Freight charges C$150 per truck plus C$4.80 per m^3 of concentrate. Northern Cargo charges US$110 per truck plus US$0.12 per ft^3 of concentrate. Using C$1 = US$0.92 and 2.54 cm = 1 inch, and using the volume and number of trucks from parts (a) and (b), determine which company is cheaper.